Public Economics (ECON 131)
Section #11: Moral Hazard & Social Security

GSI: Steve Ramos

April 14, 2025

from last week: asymmetric information as market failure

- **Adverse selection**: individuals with higher risk are more likely to purchase insurance. This makes insurance costlier, and may make the market unravel
- **Moral hazard**: individuals may change their behavior in response to being insured

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1 Moral Hazard

1.1 Practice Problem

A firm has to decide an insurance plan to provide for a single worker. The worker has probability q equal to 0.2 of becoming unemployed, and $1-q$ equal to 0.8 of staying employed. The worker can choose his unemployment duration, D, to be maximum 1 year, i.e. $D \in [0,1]$.

If the worker stays employed he gets an income of $w = 100$ minus the insurance premium p. If he loses his job he gets the insurance benefit bD for the part of the year that he is unemployed, and $w(1-D) = 100(1-D)$ for the part of the year that he is employed. In addition, he gets disutility from searching for a job. Since searching less for a job implies longer unemployment durations, this is the same as saying that he gets positive utility from a longer unemployment duration, D. With an instantaneous utility function $u(c) = ln(c)$ this means that expected utility is

benefit Δ depending on duration

$$EU = 0.8ln(100-p) + 0.2(ln(bD + 100(1-D) - p) + D)$$

where $ln(100-p)$ is his utility under employment and $ln(bD + 100(1-D) - p) + D$ is his utility under unemployment.

doesn't like searching for a job (disutility)

↓

don't search and duration of unemployment, D, gets longer

$bD + 100 - 100D - P$ → premium

benefits, income, costs

(a) What is the benefit level as a function of the premium for an actuarially fair insurance program?

$EP = p - 0.2bD = 0 \Rightarrow p = 0.2bD$

$b = \frac{p}{0.2D}$

(b) The insurance company expects that the worker will be unemployed for half a year on average, i.e. $D = 0.5$. Which premium and benefit level ensures full insurance under the actuarially fair insurance program in the previous question? Call this Policy #1.

$EU = 0.8\ln(100 - 0.2bD) + 0.2[\ln(bD + 100(1-D) - 0.2bD) + D]$

and derive w.r.t. b

$\frac{\partial EU}{\partial b} = \frac{0.8}{100 - 0.2bD} \cdot (-0.2D) + \frac{0.2}{bD + 100(1-D) - 0.2bD} \cdot (\overbrace{D - 0.2D}^{0.8D}) = 0$

cancel out

$\frac{0.8 \times 0.2D}{100 - 0.2bD} = \frac{0.8 \times 0.2D}{bD + 100(1-D) - 0.2bD}$

$bD + 100(\cancel{1} - D) - \cancel{0.2bD} = \cancel{100} - \cancel{0.2bD}$

$bD - 100D = 0 \Rightarrow bD = 100D \Rightarrow b = 100$

$p = 0.2(100)(0.5)$

$p = 10$

under full insurance

consumption smoothing

employed unemployed

$\cancel{100} - \cancel{0.2bD} = bD + 100(\cancel{1} - D) - \cancel{0.2bD}$

$0 = bD - 100D$

$100D = bD \Rightarrow b = 100$

$p = 0.2(100)(0.5)$

$p = 10$

(c) It turns out that the insurance company's guess about the unemployment duration length might not have been correct. Write up an expression for the worker's utility under Policy #1. Which D will the worker actually choose? What is his utility under Policy #1?

$EU = 0.8\ln(100 - 10) + 0.2[\ln(100D + 100(1-D) - 10) + D]$

$= 0.8\ln(90) + 0.2\ln(90) + 0.2D$

$= \ln(90) + 0.2D$

to figure out what level of D will maximize EU, take $\frac{\partial EU}{\partial D}$

here, we know $D \in [0, 1]$

if $D = 1$, EU will be the highest

$EU = \ln(90) + 0.2(1) = 4.6998$

if $D = 0.5$, EU would be

$EU = \ln(90) + 0.2(0.5) = 4.5998$

(d) The firm learns that it was losing money under Policy #1. Hence, it proposes a new actuarially fair insurance plan with full insurance under the actual D. Call this Policy #2. What is the new premium p and benefit b? What is the worker's utility under this policy? Compare it to his utility under Policy #1. Would he choose to change his unemployment duration rather than keep it at $D = 0.5$?

under full insurance, consumption when employed equals consumption when unemployed

$$\cancel{100} - \cancel{0.2bD} = bD + \cancel{100}(\cancel{1} - D) - \cancel{0.2bD}$$

$$\Rightarrow 100D = bD \Rightarrow b = 100$$ doesn't change!

actuarially fair insurance premium: $p = 0.2bD \Rightarrow 0.2(100)(1) = 20$

$$EU = 0.8\ln(100-20) + 0.2\ln(100 + 100(1-1) - 20) + 0.2(1)$$

$$= 0.8\ln(80) + 0.2\ln(80) + 0.2$$

$$= \ln(80) + 0.2$$

$$= 4.58$$

$\longrightarrow$ would have rather kept duration 0.5 w/ $b = 100$ and $p = 10$

(e) Now, suppose the firm implements a new plan (Policy #3) under which full insurance is not provided. In particular the firm provides a benefit of $b = 20$. Under this lower benefit the firm expects a low unemployment duration, namely $D = 0.25$. Under an actuarially fair program this means that $p = 1$ (check this!).

$p = 0.2(20)(0.25) = 1$ ✓

(i) Which unemployment duration will the worker actually choose?

(ii) Will the firm lose money under this D?

(iii) Compute the worker's utility. Which of the three policies does he ~~provide~~ prefer!?

(iv) Why does your answer differ from the result in class that full insurance is optimal under no behavioral responses?

(i) worker's $EV = 0.8\ln(100-1) + 0.2\ln(20D + 100(1-D) - 1) + 0.2D$

$= 0.8\ln(99) + 0.2\ln(99 - 80D) + 0.2D$

$\frac{\partial EV}{\partial D} = \frac{0.2}{99-80D}(-80) + 0.2 = 0 \Rightarrow \frac{80}{99-80D} = 1 \Rightarrow 80 = 99 - 80D$

$\Rightarrow 80D = 19$

$D = \frac{19}{80} = 0.2375$

lose \$\$ if

(ii) $p - 0.2bD < 0 \Rightarrow p < 0.2bD$

$\Rightarrow 1 < 0.2(20)D$

$\Rightarrow \frac{1}{4} < D$

$\Rightarrow 0.25 < D$

but $\frac{19}{80} < 0.25$, so they make a profit!

(iii) $EV = 0.8\ln(99) + 0.2\ln\left(99 - 80\left[\frac{19}{80}\right]\right) + 0.2\left[\frac{19}{80}\right]$

$= 0.8\ln(99) + 0.2\ln(80) + 0.2\left[\frac{19}{80}\right]$

$= 4.6$

he will choose policy #3. under full insurance, cannot commit to keep short D, so to break even, insurance will offer higher premium.

under less-than-full insurance can commit w/ cheap insurance.

2 Social Security

2.1 Definition and Motivation

- In the standard **life-cycle model**, rational individuals save while they work and consume savings when they retire
  - Uncertainty in returns to saving, life span, future ability to work, future health
  - Elderly would like *annuities*: guaranteed stream of income until end-of-life
- Rationales for govt intervention: individual failures, adverse selection in annuities market
  - For 2/3 of retirees, SS is > 50% of retirement *income* (≈ 30%: >90% of income)
  - Retirees also have *wealth* (home ownership, employment pensions, extra savings), but often insufficient
- Government intervention to provide retirement programs has been common since the first part of the 20th century
  - **Funded Programs:** Workers contributions are invested in financial assets and will pay for benefits when they retire.
  - **Unfunded Program:** Benefits of current retirees are paid out of contributions from current workers.
- US Social Security's OASI is a mandated, unfunded annuity run by the federal government

2.2 Details

- Eligibility:
  - Funded by payroll taxes on workers (12.4% on first $137,700 of earnings)
  - Need 10 years of contributions
- Payouts/benefits:
  - Nonlinear function of average earnings over highest 35 years ($1400/month average)
  - Spouses can receive 50% of higher-earner's benefit ("spousal benefit")
  - Individuals can "claim" anytime after age 62, but benefits increase with your age at claiming (actuarially fair on average - severe distributional issues)
  - Wording is important/intentional: "reduced" benefits at 62-64 (**Early** Eligibility Age), "full" benefits at 65 (**Normal**/**Full** Benefits Age), mandated payments at 70
  - Beneficiaries are expected to be mostly-retired: increases in labor income reduce current benefits (beyond a small disregard; about 50% reduction on average)
  - **Social Security Wealth:** Expected PDV of Benefits - Expected PDV of payroll tax payments

2.3 Redistribution Practice Problem

What are the redistributive effects of social security?

- across generations: redistribution from young to old
- w/in generations: redistribution from high- to low-income

2.4 Math Practice Problem

Consider the following model of social security. N people are born each period. Each person lives for two periods. In the first period of life, a person is young and in the second old. Thus, in any period after the initial period, half the population is young and half is old. Young people earn 2 chocolate bars while old people earn nothing. Assume that the chocolate melts, so there is no way for people to save privately.

Give a brief rationale for government provision of social security in this model.

individuals cannot save privately b/c chocolate melts so the gov't must provide social security so they can eat when older

Now suppose that the government of Candyland implements a social security system in the following manner. The government taxes each young person 1 chocolate bar and redistributes it to an old person in the same period. The program starts between periods 0 and 1 and ends between periods 2 and 3, as shown below. Let c^g_A denote the chocolate consumption of an agent from generation g (generation refers to the period the agent is born) at age A (age is either young or old).

(a) Under Candyland's social security program, fill in the blanks in the following chart:

Period 0		Period 1	Period 2		Period 3
	Start SS			End SS	
$c^0_{young} =$ 2		$c^0_{old} =$ 1			
		$c^1_{young} =$ 1	$c^1_{old} =$ 1		
			$c^2_{young} =$ 1		$c^2_{old} =$ 0

Which generation gains the most in terms of consumption? The least?

c^0 never pay tax but get the benefit

c^2 pays tax but never gets benefit

(b) What assumption does this model make about the effect of social security provision on retirement behavior? Discuss how empirical evidence on retirement decisions and social security relates to this assumption.

→ assumes retirement behavior is not affected by the provision of SS

→ workers' decision to retire are heavily influenced by whether or not additional work will increase SS benefits

(c) The country of Twixland is also considering implementing social security. However, the ingenious residents of Twixland have figured out a way to freeze and save chocolate for retirement. All of them have utility over consumption when young (c_A) and old (c_B) given by $\log(c_A) + \log(c_B)$. The demographic and other aspects of the economy are as in Candyland.

concave smooth consumption!

Replicate the chart in (b) for Twixland, assuming that the start and end of SS are completely unanticipated by its residents.

one saved, one SS

Period 0		Period 1	Period 2		Period 3
	Start SS			End SS	
$c^0_{young} = 1$	|	$c^0_{old} = 2$		|	
	|	$c^1_{young} = 1$	$c^1_{old} = 1$	|	
	|		$c^2_{young} = 1$	|	$c^2_{old} = 0$

again

c^0 never pay tax but get the benefit

↓

consume more when old now before consumped more more when young (a)

c^2 pays tax but never gets benefit